Experiment No:12

Aim

To perform a **Perspective Transformation** on an image using Python and OpenCV.

Procedure

1. Import the required libraries (cv2 and numpy).
2. Read the input image.
3. Define four source points and four destination points.
4. Compute the perspective transformation matrix using cv2.getPerspectiveTransform().
5. Apply the transformation using cv2.warpPerspective().
6. Display the original and transformed images.
7. Close all windows after a key is pressed.

Code

```
import cv2
import numpy as np
img = cv2.imread("image.jpg")
if img is None:
    print("Error: Image not found!")
    exit()
rows, cols = img.shape[:2]
pts1 = np.float32([[50, 50], [300, 50], [50, 300], [300, 300]])
pts2 = np.float32([[0, 0], [300, 50], [50, 300], [300, 300]])
M = cv2.getPerspectiveTransform(pts1, pts2)
perspective_img = cv2.warpPerspective(img, M, (cols, rows))
cv2.imshow("Original Image", img)
cv2.imshow("Perspective Transformed Image", perspective_img)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

Sample Input:

Output:

Result

The **Perspective Transformation** was successfully performed on the input image. The transformed image showed a change in viewpoint, making the image appear tilted or viewed from a different angle while preserving straight lines.